



Vibration suppression characteristics of piezoelectric metamaterial pipe conveying fluid based on shunt circuits

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ABSTRACT

This paper proposes a novel metamaterial pipe with controllable bandgap characteristics, which is constructed by attaching shunted piezoelectric patches to a fluid-conveying pipe along the axial direction. The vibration transmission coefficient and wave attenuation constant of the metamaterial pipe are calculated using an analytical model that combines Euler-Bernoulli beam theory and Timoshenko beam theory. The main contribution of this work lies in the combined analytical model can calculate the vibration and wave propagation characteristics of circular metamaterial pipe with sectorial annular piezoelectric patches. The calculation accuracy is quantitatively verified through the results in published literature and finite element numerical simulations. Research results indicate that Bragg scattering (BS) bandgap and electromechanical local resonance (LR) bandgap can be generated and used for low-frequency vibration reduction. Due to the damping effect of internal fluid flow, the weak non-reciprocal propagation of bending waves occurs for the metamaterial pipe with non-zero fluid velocity. The transmission coefficient can be tuned by flow velocity, and controllable bandgap characteristics can be achieved by designing shunt circuit parameters. In addition, the bandgap coupling characteristics of metamaterial pipes are studied and the hybrid bandgap is obtained, which can be used to achieve broadband low-frequency vibration reduction of engineering fluid-conveying pipes.

1. Introduction

Fluid-conveying pipes play a significant role in petroleum and petrochemical engineering applications, such as piping systems in chemical plants, above-ground natural gas conveyance pipes, flexible marine risers and so forth (Ibrahim, 2010; Païdoussis, 2022; Zhang et al., 2022). The dynamic characteristics of fluid-conveying pipes can be affected by internal fluid (Guo et al., 2022; Lee and Park, 2006), especially when the vibration problems are encountered due to internal fluid-induced forces or external mechanical excitations. Therefore, the vibration and wave propagation characteristics of fluid-conveying pipes have been attracting substantial attention, and different suppression techniques were developed for diverse pipe systems (Ding et al., 2019; Chen et al., 2021; Pisarski et al., 2018).

Transfer matrix method was used to analyze the vibration of fluid-conveying pipe (Li et al., 2014), and the results showed that the

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vibration reduction performance can be improved by selecting optimal system parameters. The bandgap characteristics of fluid-conveying pipes supported by equidistantly supports were explored (Koo and Park, 1998). It was found that the vibration reduction of piping system could be achieved by designing the periodic supports. Subsequently, different pipe structures with bandgap feature were presented to suppress specific frequency mechanical vibrations based on the idea of phononic crystal or metamaterial (Liang and Qian, 2024; Ruzzene and Baz, 2001). The periodic pipe with alternating materials in radial and axial directions was designed, and it was found that lower frequency bandgap or wider bandgap can be obtained (Zhang et al., 2017). The local resonance (LR) bandgap mechanism was also introduced to construct periodic pipe structures (Iqbal et al., 2020), and desired low-frequency bandgap could be realized by designing the local resonators. However, the LR bandgap usually occurs over a narrow frequency range, which limits its practical applications in engineering vibration reduction. Therefore, numerous researches on bandgap enhancement have been carried out to achieve broadband vibration reduction (Li et al., 2025; Dal'Poggetto et al., 2021), and new methods with potential application value in pipe vibration control have also been developed (Dong et al., 2024; Zhao et al., 2025; Tian et al., 2025).

In order to meet the vibration reduction requirements in various application scenarios, controllable bandgap property become particularly popular (Chen et al., 2017; Doki and Hiramoto, 1998; Wang et al., 2020). In traditional bandgap design, the width and position of bandgaps are usually customized by rational selection of material and structural parameters. Piezoelectric materials have been adopted in the design of active metamaterial structures due to the excellent tunable vibration reduction performance (Ma et al., 2021; Kherraz et al., 2018). By exploiting periodically distributed piezoelectric patches, an active metamaterial beam with controllable bending stiffness was designed (Yi et al., 2019). This research showed that the width and location of BS bandgaps could be tuned by using active control strategy. The formation of electromechanical LR bandgap in piezoelectric bi-morph metamaterial plate with segmented electrodes was investigated (Sugino et al., 2018). It was shown that the wave propagation characteristics of LR bandgaps depended on the piezoelectric material and resonant circuit parameters. In addition, many other piezoelectric-based active metamaterials have also been designed and studied to generate controllable BS and LR bandgaps (Alroldi and Ruzzene, 2011; Chen et al., 2013).

Due to the special mechanical properties of non-reciprocal metamaterials, the application in unidirectional devices is becoming increasingly attractive (Nassar et al., 2020; Gao and Wang, 2022). In non-reciprocal mechanical systems, wave-like excitations can propagate with different amplitudes in one direction and the opposite (Palermo et al., 2020). Over the past few decades, extensive efforts have been devoted to break the reciprocity (Brandenbourger et al., 2019; Blanchard et al., 2018). By exploiting strong nonlinear or time-dependent material properties, the non-reciprocal effects could be achieved (Moore et al., 2018). A compact acoustic circulator was proposed by introducing a circulating fluid to form a mechanical bias (Fleury et al., 2014), and the weak non-reciprocal wave propagation was achieved in the resonant ring cavity. Although the flow damping of fluid-conveying pipes has been studied previously (Païdoussis et al., 2008), the in-depth exploration of non-reciprocity for metamaterial pipe biased by internal fluid has not been carried out.

At present, the vibration problems of fluid-conveying pipes have been widely studied (Wang and Shen, 2019; Liang et al., 2023). Although the vibration suppression of phononic crystal pipe conveying fluid has been investigated, the pipe structure is considered to

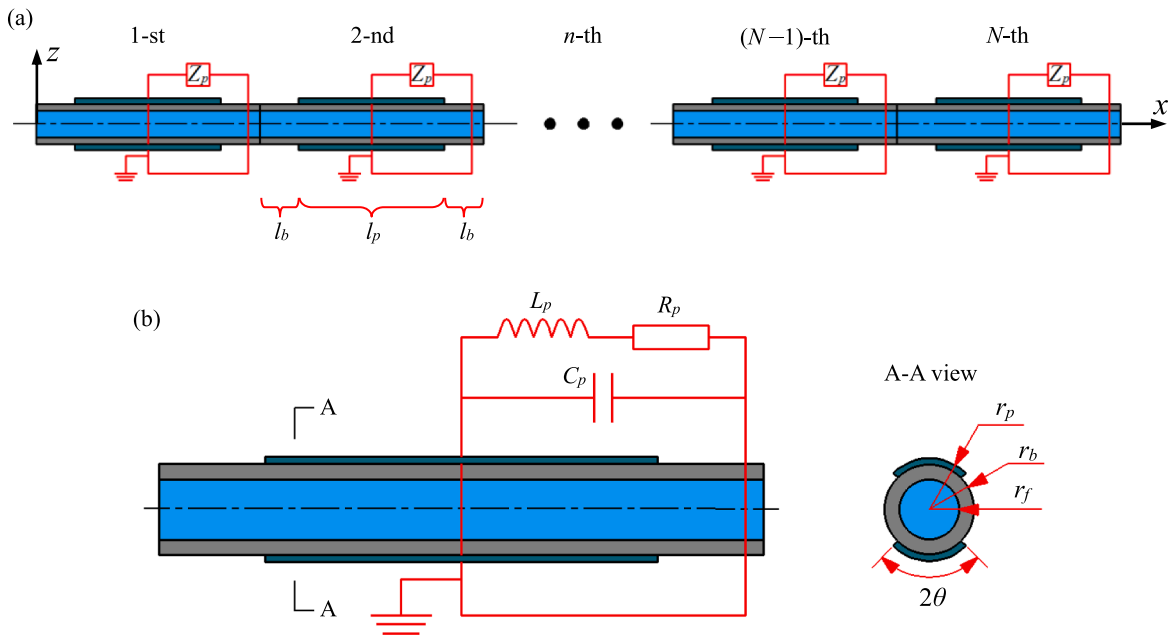


Fig. 1. Schematics of metamaterial pipe: (a) piezoelectric metamaterial pipe with N unit-cells, (b) the n -th unit-cell of the piezoelectric metamaterial pipe.

have a rectangular cross-section (Lyu et al., 2020). In addition, the vibration energy harvesting for fluid-conveying pipe with circular cross-section was studied, but the piezoelectric material covered the entire out-wall of pipe structure (Lu et al., 2022). It is noteworthy that there is a lack of research on the bandgap characteristics of circular cross-sectional fluid-conveying pipes partially covered by piezoelectric patches in the circumferential direction.

Previous studies on the dynamics of fluid-conveying pipes based on different beam theories have shown that Euler-Bernoulli beam theory may not be sufficient to describe the cross-sectional shear effects of flexible short pipes and high-order modes in long pipes (Liu et al., 2020; Li et al., 2024; Miranda and Dos Santos, 2019). Therefore, this article proposes a combined model for calculating the dynamic characteristics of metamaterial pipes. The sectional shear effect of the host pipe is considered and the equations of motion are established based on Timoshenko beam theory. The attached piezoelectric patches are modeled using Euler-Bernoulli beam theory, so the unknown shear factor can be avoided. The novelty of this work lies in the combined analytical model can calculate the vibration and wave propagation characteristics of circular metamaterial pipe with sectorial annular piezoelectric patches.

In light of the above literature analysis, a novel metamaterial pipe is constructed by periodically attaching shunted piezoelectric patches to a fluid-conveying pipe. A combined analytical model is proposed to calculate the vibration and wave propagation characteristics of the metamaterial pipe, and the accuracy is validated by finite element simulation and existing literature results. The BS and LR bandgaps are opened in low-frequency range, and broadband vibration reduction of the metamaterial pipe can be realized based on the bandgap coupling characteristics. The influence of internal fluid velocity on wave propagation of the metamaterial pipe is explored, and the tunable bandgap characteristics of the metamaterial pipe are achieved by adjusting the shunt circuits. This study may provide some valuable results for the vibration reduction design of fluid-conveying pipe systems.

2. Description of the metamaterial pipe

The metamaterial pipe is composed of N unit-cells as displayed in Fig. 1(a), and the n -th representative unit-cell is further illustrated in Fig. 1(b). The unit-cell contains three parts: (i) the host pipe conveying fluid, (ii) the shunted piezoelectric patches and (iii) the external shunting circuits. The upper and lower piezoelectric patches have same structural dimensions, and the piezoelectric patches are polarized outward along the radial direction. The unit-cell of metamaterial pipe can be divided into three segments or sub-cells, such as the middle segment with shunted piezoelectric patches and the two bare pipe segments at its left and right ends.

The three segments are considered as sub-cells that make up the unit-cell, where the length of the two bare pipe segments is l_b , and the length of the pipe with shunted piezoelectric patches is l_p . Then, the total length of the unit-cell can be expressed as $l_a = (l_p + 2l_b)$, i. e. the sum of the lengths of three sub-cells displayed in Fig. 1(a). The inner and outer radii of the metamaterial pipe are represented by r_f and r_b . The piezoelectric patches are mounted on the outer wall of the host pipe, so that the inner radius of piezoelectric patch is equal to the outer radius of host pipe, and the outer radius of piezoelectric patch is denoted by r_p . The spanning angle of piezoelectric patch is 2θ . In each external shunting circuit, the resistance R_p and inductance L_p are connected in series and then in parallel with the capacitance C_p . The impedance of each external shunting circuit is denoted by Z_p , which can be designed to achieve desired vibration reduction performance.

The vibration of fluid-conveying pipes under finite deformation is considered, where bending vibration and axial vibration can be decoupled, and the bending vibration is more easily excited in practical engineering. Therefore, we focus on the bending vibration reduction of fluid-conveying pipes. The bending vibration equation with respect to the deflection $w(x, t)$ will be established based on the x - z coordinate system. Among them, x is the distance along the axial coordinate, and t denotes time.

3. Analytical model of the metamaterial pipe

In this section, an analytical model is established for the proposed metamaterial pipe. Timoshenko beam theory is adopted to describe the motion of the host pipe, while the piezoelectric patches are modeled using Euler-Bernoulli beam theory to avoid the unknown sectional shear factor. Subsequently, the dynamic stiffness equation of the metamaterial pipe is derived to calculate the transmission coefficient and attenuation constant, which can be used to analyze the vibration transmission and wave propagation characteristics of the metamaterial pipe.

3.1. Model formulation

According to Timoshenko beam theory, the axial stress and shear stress of the host pipe are considered for the metamaterial pipe, and the constitutive relation can be written as

$$\begin{Bmatrix} \sigma_1 \\ \tau_3 \end{Bmatrix} = \begin{bmatrix} Y_b & 0 \\ 0 & G \end{bmatrix} \begin{Bmatrix} \varepsilon_1 \\ \gamma_3 \end{Bmatrix} \quad (1)$$

where, σ_1 and τ_3 are axial and shear stresses, ε_1 and γ_3 are axial and shear strains, Y_b and G are Young's modulus and shear modulus of the host pipe.

For the piezoelectric patches, the piezoelectric constitutive equation based on Euler-Bernoulli beam theory can be obtained by neglecting the stress components other than the stress in x -direction, and it is given by

$$\begin{Bmatrix} \sigma_1^p \\ D_3 \end{Bmatrix} = \begin{bmatrix} c_{11}^E & -e_{31} \\ e_{31} & \epsilon_{33}^S \end{bmatrix} \begin{Bmatrix} S_1 \\ E_3 \end{Bmatrix} \quad (2)$$

where, σ_1^p and S_1 represent axial stress and strain of the piezoelectric patch, D_3 and E_3 are electric displacement and electric intensity, c_{11}^E denotes the elastic coefficient, e_{31} is the piezoelectric coupling coefficient, and ϵ_{33}^S indicates the dielectric coefficient for constant strain.

Timoshenko point-force plug-flow theory has been obtained (Paidoussis and Laithier, 1976; Paidoussis, 2014), where the equations of motion were derived by Newtonian methods, and the plug-flow model was retained. The formulations of the host pipe conveying fluid can be obtained by omitting the terms relevant to gravity, and the formulations of piezoelectric patch based on Euler-Bernoulli beam theory can be derived by omitting the shear effect and rotary inertia. Then, the equations of motion of the piezoelectric metamaterial pipe under short circuit condition can be expressed in x - z coordinate system as the following unified form:

$$(\rho A)_{eff} \frac{\partial^2 w(x, t)}{\partial t^2} + 2\rho_f A_f v \frac{\partial^2 w(x, t)}{\partial x \partial t} + \rho_f A_f v^2 \frac{\partial^2 w(x, t)}{\partial x^2} - \kappa G A_b \left[\frac{\partial^2 w(x, t)}{\partial x^2} - \frac{\partial \varphi(x, t)}{\partial x} \right] = 0, \quad (3)$$

$$(YI)_{eff} \frac{\partial^2 \varphi(x, t)}{\partial x^2} + \kappa G A_b \left[\frac{\partial w(x, t)}{\partial x} - \varphi(x, t) \right] - (\rho_b I_b + \rho_f I_f) \frac{\partial^2 \varphi(x, t)}{\partial t^2} = 0 \quad (4)$$

where, $w(x, t)$ and $\varphi(x, t)$ are transverse displacement and rotation angle, $(YI)_{eff}$ and $(\rho A)_{eff}$ are effective bending stiffness and effective mass of the metamaterial pipe, v denotes the internal fluid velocity, κ is shear coefficient of the pipe structure, ρ and A are mass density and area of the cross section, the subscripts b , p and f represent the host pipe, piezoelectric patch and internal fluid. For the n -th unit-cell of the metamaterial pipe, the effective stiffness $(YI)_{eff}$ and effective mass $(\rho A)_{eff}$ are given by

$$(YI)_{eff} = \begin{cases} \frac{1}{4} Y_b \pi (r_b^4 - r_f^4), & x_n \in [(n-1)l_a, (n-1)l_a + l_b] \cup [nl_a - l_b, nl_a] \\ \frac{1}{4} [Y_b \pi (r_b^4 - r_f^4) + c_{11}^E (r_p^4 - r_b^4) (2\theta + \sin 2\theta)], & x_n \in [(n-1)l_a + l_b, nl_a - l_b] \end{cases}, \quad (5)$$

$$(\rho A)_{eff} = \begin{cases} \rho_b A_b + \rho_f A_f, & x_n \in [(n-1)l_a, (n-1)l_a + l_b] \cup [nl_a - l_b, nl_a] \\ \rho_b A_b + 2\rho_p A_p + \rho_f A_f, & x_n \in [(n-1)l_a + l_b, nl_a - l_b] \end{cases}. \quad (6)$$

The upper and lower piezoelectric patches are connected in series, so the total electric charge of the electrodes of the piezoelectric patches can be expressed as

$$Q_{total}^s = \int_{top} \int D_3 dA_s = \int_{bottom} \int D_3 dA_s \quad (7)$$

where, Q_{total}^s is the generated electric charge due to the piezoelectric effect, A_s is the area of the electrodes.

According to the Kirchhoff's current law, the output current formed by the electric charge of the piezoelectric patch should be equal to the current in external shunting circuit. Therefore, we have

$$I(t) = \frac{dQ_{total}^s(t)}{dt} = \frac{u(t)}{Z_p} \quad (8)$$

where, $I(t)$ and $u(t)$ are current and voltage of the circuit. Substituting the Eq. (7) into Eq. (8), the circuit equation can be obtained and given as follows:

$$C_p^s \frac{du(t)}{dt} + \frac{u(t)}{Z_p} + \vartheta \frac{\partial^2 w(l_p, t) - \partial^2 w(0, t)}{\partial t \partial x} = 0 \quad (9)$$

where, C_p^s is the inherent capacitance of the piezoelectric patch, ϑ represents the electromechanical coupling coefficient, and their expressions are given as follows:

$$C_p^s = \frac{1}{4} \epsilon_{33}^S l_p \theta \frac{r_p + r_b}{r_p - r_b}, \quad (10a)$$

$$\vartheta = \frac{2}{3} d_{31} c_{11}^E (r_b^2 + r_p^2 + r_b r_p) \sin \theta. \quad (10b)$$

The shear force and bending moment of the metamaterial pipe can be derived as

$$V(x, t) = \kappa G A_b \left[\frac{\partial w(x, t)}{\partial x} - \varphi(x, t) \right], \quad (11)$$

$$M(x, t) = -(YI)_{eff} \frac{\partial^2 w(x, t)}{\partial x^2} + \vartheta u(t) [H(x - x_n^L) - H(x - x_n^R)] \quad (12)$$

where, $H(\cdot)$ is the Heaviside function, x_n^L and x_n^R represent the left and right edges of the piezoelectric patch in n -th unit cell.

3.2. Dynamic stiffness equation

By performing Fourier transformation on the dynamic equations in Eqs. (3) and (4), the wave solutions of displacement filed for the metamaterial pipe can be assumed as

$$W(x) = \sum_{j=1}^4 \alpha_j e^{-i\lambda_j x}, \quad (13a)$$

$$\Phi(x) = \sum_{j=1}^4 \beta_j \alpha_j e^{-i\lambda_j x} \quad (13b)$$

where, $W(x)$ and $\Phi(x)$ are transverse displacement and rotation angle in frequency domain, i is the imaginary unit, α_j and β_j ($j = 1, 2, 3, 4$) are the wave amplitude coefficients, and λ_j ($j = 1, 2, 3, 4$) signify the characteristic wave numbers, which can be determined by the characteristic wave equation:

$$c_1 \lambda_j^4 + c_2 \lambda_j^3 + c_3 \lambda_j^2 + c_4 \lambda_j + c_5 = 0 \quad (14)$$

where, the coefficients of Eq. (14) are given as follows:

$$c_1 = \rho_f A_f (YI)_{eff} v^2 - (YI)_{eff} \kappa G A_b, \quad (15a)$$

$$c_2 = -2\rho_f A_f (YI)_{eff} v \omega, \quad (15b)$$

$$c_3 = \rho_f A_f \kappa G A_b v^2 + (YI)_{eff} (\rho A)_{eff} \omega^2 - \rho_f A_f (\rho_b I_b + \rho_f I_f) v^2 \omega^2 + (\rho_b I_b + \rho_f I_f) \kappa G A_b \omega^2, \quad (15c)$$

$$c_4 = 2\rho_f A_f (\rho_b I_b + \rho_f I_f) v \omega^3 - 2\rho_f A_f \kappa G A_b v \omega, \quad (15d)$$

$$c_5 = \kappa G A_b (\rho A)_{eff} \omega^2 - (\rho A)_{eff} (\rho_b I_b + \rho_f I_f) \omega^4. \quad (15e)$$

Then, the wave numbers λ_j ($j = 1, 2, 3, 4$) can be obtained by solving Eq. (14), and the wave amplitude coefficients β_j ($j = 1, 2, 3, 4$) can be obtained as

$$\beta_j = \frac{\kappa G A_b \lambda_j^2 - (\rho A)_{eff} \omega^2}{\kappa G A_b i \lambda_j}, (j = 1, 2, 3, 4). \quad (16)$$

With the help of Eq. (13) and assuming the harmonic solution $u(t) = U(\omega) e^{i\omega t}$, the expressions of shear force and bending moment in Eqs. (11) and (12) can be expressed in frequency domain as

$$\bar{V}(x) = \kappa G A_b \left[\frac{\partial W(x)}{\partial x} - \Phi(x) \right], \quad (17)$$

$$\bar{M}(x) = (YI)_{eff} \frac{\partial^2 W(x)}{\partial x^2} - \bar{M}_p [H(x - x_n^L) - H(x - x_n^R)] \quad (18)$$

where, $\bar{M}_p$ is actuation moment generated by the piezoelectric patch, which can be obtained by using the current balance equation in Eq. (9), and it is given by

$$\bar{M}_{pj} = \frac{i\omega \vartheta^2 Z_p}{(1 + i\omega C_p^s Z_p)} \frac{\partial}{\partial x} [W(l_p) - W(0)], (j = 1, 2, 3, 4). \quad (19)$$

For each unit-cell of the metamaterial pipe, the displacement and force boundary conditions of the sub-cells are given by

$$W_1 = W(0), \quad (20a)$$

$$\Phi_1 = \Phi(0), \quad (20b)$$

$$W_2 = W(l_{n,k}), \quad (20c)$$

$$\Phi_2 = \Phi(l_{n,k}), \quad (20d)$$

$$\bar{V}_1 = \bar{V}(0), \quad (21a)$$

$$\bar{M}_1 = \bar{M}(0), \quad (21b)$$

$$\bar{V}_2 = \bar{V}(l_{n,k}), \quad (21c)$$

$$\bar{M}_2 = \bar{M}(l_{n,k}) \quad (21d)$$

where, $l_{n,k}$ represents the length of the k -th sub-cell in n -th unit-cell, the subscripts 1 and 2 represent the left and right nodes of the sub-cells.

Substituting Eq. (13) into Eq. (20), the displacement vector can be expressed as

$$\mathbf{d}_{n,k} = \mathbf{Q}_{n,k} \boldsymbol{\alpha} \quad (22)$$

where, $\mathbf{d}_{n,k} = [W_1, \Phi_1, W_2, \Phi_2]$ is the nodal displacement vector of the sub-cell, $\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \alpha_3, \alpha_4]$ is the wave coefficient vector, the specific form of matrix $\mathbf{Q}_{n,k}$ is given as follows:

$$\mathbf{Q}_{n,k} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ \beta_1 & \beta_2 & \beta_3 & \beta_4 \\ e^{-i\lambda_1 l_{n,k}} & e^{-i\lambda_2 l_{n,k}} & e^{-i\lambda_3 l_{n,k}} & e^{-i\lambda_4 l_{n,k}} \\ \lambda_1 e^{-i\lambda_1 l_{n,k}} & \lambda_2 e^{-i\lambda_2 l_{n,k}} & \lambda_3 e^{-i\lambda_3 l_{n,k}} & \lambda_4 e^{-i\lambda_4 l_{n,k}} \end{bmatrix}. \quad (23)$$

Substituting Eqs. (17)–(18) into Eq. (21), the force vector can be obtained in the same form as that in Eq. (22), and it is written as

$$\mathbf{f}_{n,k} = \mathbf{P}_{n,k} \boldsymbol{\alpha} \quad (24)$$

where, $\mathbf{f}_{n,k} = [V_1, M_1, V_2, M_2]$ is the nodal force vector of the sub-cell, and the detailed expression of matrix $\mathbf{P}_{n,k}$ is given as follows:

$$\mathbf{P}_{n,k} = \begin{bmatrix} \kappa G A_b (i\lambda_1 + \beta_1) & \kappa G A_b (i\lambda_2 + \beta_2) & \kappa G A_b (i\lambda_3 + \beta_3) & \kappa G A_b (i\lambda_4 + \beta_4) \\ (YI)_{eff} i\lambda_1 \beta_1 - \bar{M}_{p,1} & (YI)_{eff} i\lambda_2 \beta_2 - \bar{M}_{p,2} & (YI)_{eff} i\lambda_3 \beta_3 - \bar{M}_{p,3} & (YI)_{eff} i\lambda_4 \beta_4 - \bar{M}_{p,4} \\ -\kappa G A_b (i\lambda_1 + \beta_1) e^{-i\lambda_1 l_{n,k}} & -\kappa G A_b (i\lambda_2 + \beta_2) e^{-i\lambda_2 l_{n,k}} & -\kappa G A_b (i\lambda_3 + \beta_3) e^{-i\lambda_3 l_{n,k}} & -\kappa G A_b (i\lambda_4 + \beta_4) e^{-i\lambda_4 l_{n,k}} \\ -(YI)_{eff} i\lambda_1 \beta_1 e^{-i\lambda_1 l_{n,k}} + \bar{M}_{p,1} & -(YI)_{eff} i\lambda_2 \beta_2 e^{-i\lambda_2 l_{n,k}} + \bar{M}_{p,2} & -(YI)_{eff} i\lambda_3 \beta_3 e^{-i\lambda_3 l_{n,k}} + \bar{M}_{p,3} & -(YI)_{eff} i\lambda_4 \beta_4 e^{-i\lambda_4 l_{n,k}} + \bar{M}_{p,4} \end{bmatrix}. \quad (25)$$

Combining Eqs. (22) and (24), the dynamic stiffness equation of the sub-cell can be obtained as

$$\mathbf{f}_{n,k} = \mathbf{K}_{n,k} \mathbf{d}_{n,k} \quad (26)$$

where, $\mathbf{K}_{n,k} = \mathbf{P}_{n,k} (\mathbf{Q}_{n,k})^{-1}$ is the dynamic stiffness matrix, which gives the relations between nodal displacements and nodal forces of the sub-cells.

3.3. Vibration transmission property

There are N unit-cells in the metamaterial pipe as shown in Fig. 1(a), and the dynamic stiffness matrix of the sub-cell has been given in Eq. (26). By assembling Eq. (26) with respect to each sub-cells, the dynamic stiffness equation of the finite metamaterial pipe can be obtained as

$$\mathbf{f}_g = \mathbf{K}_g \mathbf{d}_g \quad (27)$$

where, $\mathbf{f}_g$ and $\mathbf{d}_g$ are the global force and displacement vectors, and $\mathbf{K}_g$ is the global stiffness matrix with the following form

$$\mathbf{K}_g = \begin{bmatrix} \mathbf{K}_{1,1}^{LL} & \mathbf{K}_{1,1}^{LR} & \mathbf{0} & \mathbf{0} & \cdots & \cdots & \mathbf{0} & \mathbf{0} \\ \mathbf{K}_{1,1}^{RL} & \mathbf{K}_{1,1}^{RR} + \mathbf{K}_{1,2}^{LL} & \mathbf{K}_{1,2}^{LR} & \mathbf{0} & \vdots & \vdots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{K}_{1,2}^{RL} & \mathbf{K}_{1,2}^{RR} + \mathbf{K}_{1,3}^{LL} & \cdots & \mathbf{0} & \vdots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \vdots & \ddots & \cdots & \mathbf{0} & \vdots & \vdots \\ \vdots & \vdots & \mathbf{0} & \vdots & \mathbf{K}_{N-1,3}^{RR} + \mathbf{K}_{N,1}^{LL} & \mathbf{K}_{N,1}^{LR} & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \vdots & \mathbf{0} & \mathbf{K}_{N,1}^{RL} & \mathbf{K}_{N,1}^{RR} + \mathbf{K}_{N,2}^{LL} & \mathbf{K}_{N,2}^{LR} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \vdots & \vdots & \mathbf{0} & \mathbf{K}_{N,2}^{RL} & \mathbf{K}_{N,2}^{RR} + \mathbf{K}_{N,3}^{LL} & \mathbf{K}_{N,3}^{LR} \\ \mathbf{0} & \mathbf{0} & \cdots & \cdots & \mathbf{0} & \mathbf{0} & \mathbf{K}_{N,3}^{RL} & \mathbf{K}_{N,3}^{RR} \end{bmatrix} \quad (28)$$

where, the two subscripts represent unit-cell number and sub-cell number in sequence.

In order to evaluate the vibration transmission performance of metamaterial pipe, the transmission coefficient between the excitation and response positions is introduced as

$$T_r = 20 \lg |w_{out} / w_{in}| \quad (29)$$

where, T_r is the transmission coefficient, w_{in} and w_{out} are transverse displacements at excitation and response positions, $\lg$ is a logarithmic function with a base of 10. The displacements of the metamaterial pipe can be obtained by solving Eq. (27), and the transmission coefficient can be calculated by Eq. (29). Consequently, the vibration transmission performance and bandgap characteristics of the metamaterial pipe can be analyzed.

3.4. Wave propagation property

To analyze the wave propagation characteristics of the metamaterial pipe shown in Fig. 1, the dynamic stiffness matrix in Eq. (26) is partitioned as follows:

$$\begin{Bmatrix} \mathbf{f}_L \\ \mathbf{f}_R \end{Bmatrix} = \begin{bmatrix} \mathbf{K}_{LL} & \mathbf{K}_{LR} \\ \mathbf{K}_{RL} & \mathbf{K}_{RR} \end{bmatrix} \begin{Bmatrix} \mathbf{d}_L \\ \mathbf{d}_R \end{Bmatrix} \quad (30)$$

where, $\mathbf{K}_{LL}$, $\mathbf{K}_{LR}$, $\mathbf{K}_{RL}$ and $\mathbf{K}_{RR}$ are the block matrices of dynamic stiffness matrix, and $\mathbf{f}_L$, $\mathbf{f}_R$, $\mathbf{d}_L$ and $\mathbf{d}_R$ are the force and displacement state vectors.

For the representative unit-cell in metamaterial pipe, the transfer relation between the left and right edges can be expressed as

$$\mathbf{v}_L = \prod_{k=1}^3 \mathbf{T}_k \mathbf{v}_R \quad (31)$$

where, $\mathbf{v}_L = [\mathbf{d}_L \quad \mathbf{f}_L]^T$ and $\mathbf{v}_R = [\mathbf{d}_R \quad \mathbf{f}_R]^T$ are state vectors of the left and right edges of the unit-cell, $\mathbf{T}_k$ is the transfer matrix between the left and right state vectors of the sub-cell, and it can be obtained as

$$\mathbf{T}_k = \begin{bmatrix} -(\mathbf{K}_{LR})^{-1} \mathbf{K}_{LL} & (\mathbf{K}_{LR})^{-1} \\ -\mathbf{K}_{RL} + \mathbf{K}_{RR} (\mathbf{K}_{LR})^{-1} \mathbf{K}_{LL} & -\mathbf{K}_{RR} (\mathbf{K}_{LR})^{-1} \end{bmatrix}, (k = 1, 2, 3). \quad (32)$$

Bloch theory is adopted to obtain the Bloch wave vector of the metamaterial pipe, and the relation between the left and right state vectors of the unit-cell is given by

$$\mathbf{v}_L = e^{i\mu} \mathbf{v}_R \quad (33)$$

where, μ is the Bloch wave vector used to characterize the wave propagation. The eigenvalue problem for the metamaterial pipe can be obtained by combining Eqs. (31) and (33). The imaginary part of μ is called wave attenuation constant, which represents the attenuation of bending wave in the bandgap frequency range.

4. Results and discussions

4.1. Validation of the analytical model

Prior to studying the vibration reduction performance of the metamaterial pipe, the accuracy of the proposed analytical model is verified by the finite element method (FEM). The software COMSOL Multiphysics is adopted in FEM calculation, and the effect of fluid on pipe structure is implemented using solid mechanics module. The length of the bare pipe segment is taken as $l_b = 0.06$ m, and the inner and outer radii are taken as $r_f = 0.012$ m and $r_b = 0.016$ m. The length of the pipe with piezoelectric patches is taken as $l_p = 0.18$ m, and the outer radius of piezoelectric patches is $r_p = 0.01625$ m. The span angle of each piezoelectric patch is $2\theta = 30^\circ$. The material properties of the host pipe and piezoelectric patch are listed in Table 1, and the internal fluid is considered to be water. The circuit parameters are taken as $R_p = 20 \Omega$, $L_p = 0.8$ H and $C_p = 100$ nF. The metamaterial pipe contains 12 unit-cells. For simplicity, the fluid velocity inside the pipe is set to zero. The finite metamaterial pipe is under free boundary conditions, which can better correspond to the infinite periodic metamaterials. A harmonic excitation $F = e^{i\omega t}$ is applied at the right edge, and the bending vibration response at the left edge is picked up. Then, the vibration transmission coefficient can be calculated by using Eq. (29).

In Fig. 2, the vibration and wave propagation characteristics of the metamaterial pipe are calculated using the analytical method

Table 1
Material properties of the host pipe and piezoelectric patches.

Categories	ρ (kg/m ³)	Y_b (Pa)	s_{11}^E (1/Pa)	d_{31} (C/N)	ϵ_{33}^S (F/m)
Host pipe	1180	4.35×10^9	—	—	—
Piezoelectric patches	7500	—	16.5×10^{-12}	-274×10^{-12}	3.01×10^{-8}

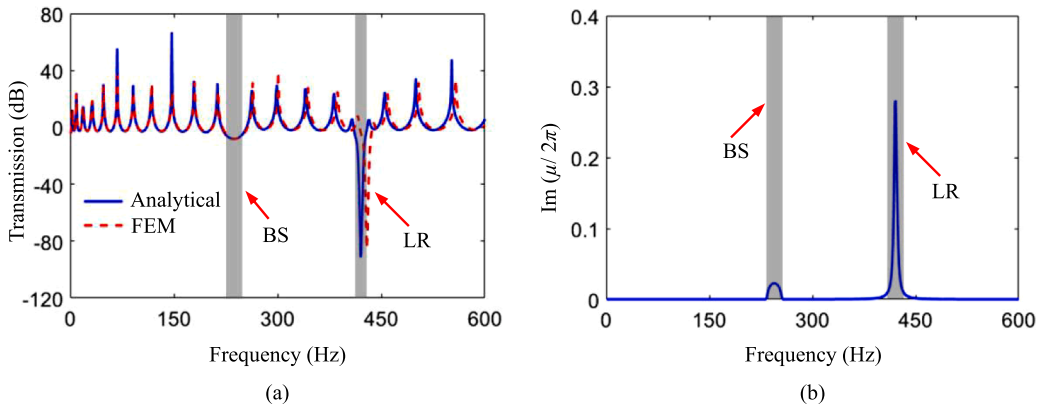


Fig. 2. Vibration and wave propagation characteristics of the metamaterial pipe: (a) transmission coefficient, (b) attenuation constant.

and FEM. It can be seen from Fig. 2(a) that the two transmission coefficients are in well agreement. Two bandgaps can be observed in the calculation frequency range, namely BS bandgap and electromechanical LR bandgap. The BS bandgap is generated due to the periodicity of metamaterial pipe, and the electromechanical LR bandgap is generated because the piezoelectric patches are shunted by external resonant shunt circuits. The maximum error occurs nearby the LR bandgap, and the frequency at the minimum transmission coefficient can be used to characterize the location of LR bandgap. The locations of LR bandgap calculated by the analytical method and FEM are 421 Hz and 428 Hz, as shown in Fig. 2(a). The calculation error between these two frequencies is about 1.64%. Therefore, the correctness of the analytical method is verified by FEM, and the calculation accuracy of the analytical model is acceptable.

The influence of fluid velocity on the vibration characteristics of metamaterial pipe is further studied by comparing the natural frequencies with those in literature (Li et al., 2013). As listed in Table 2, the proposed analytical method can accurately predict the natural frequencies under different fluid velocity. It should be indicated that when the fluid inside the of fluid-conveying pipe has a higher flow velocity, each order of the calculated natural frequencies will slightly decrease.

To illustrate the vibration reduction performance of the metamaterial pipe, the vibration distributions at different frequencies are calculated through numerical simulation, as displayed in Fig. 3. The vibration distribution of the metamaterial pipe at 249 Hz is shown in Fig. 3(a), which is located in the frequency range of BS bandgap. It can be found that the wave-like excitation can be obviously suppressed, and the vibrations can hardly propagate to the left edge. In Fig. 3(b), the vibration distributions in the frequency range of pass band are plotted at 332 Hz. In this case, the harmonic disturbance can transmit from right edge to the left without attenuation. Fig. 3(c) shows the vibration distributions at 427 Hz, which is exactly located in the frequency range of the LR bandgap. It is observed that the harmonic excitation can be radically suppressed, and the control effect of LR bandgap on vibration amplitude is much better than that of BS bandgap.

4.2. Influence of internal fluid velocity

The previous results have demonstrated that internal fluid velocity can affect the natural frequency of fluid-conveying pipe. In this section, the influence of internal fluid velocity on the dynamic response of metamaterial pipe will be examined. Two cases are considered in exploring the vibration and wave propagation characteristics. In case I, a wave-like excitation is applied on the left edge and the right edge is selected as response position. Therefore, the propagation direction of the excitation is consistent with the flow direction, as shown in Fig. 4(a). In case II, the excitation and response positions are reversed, as shown in Fig. 4(b), and the propagation direction of wave-like excitation is opposite to the flow direction. In the calculation, the values of resistance, inductance and capacitance are taken as $R_p = 50 \Omega$, $L_p = 0.8 \text{ H}$ and $C_p = 100 \text{ nF}$, respectively. The internal fluid is assumed to flow from the left end to the right, and the fluid velocity is taken as $v = 25 \text{ m/s}$.

As shown in Fig. 5, the vibration transmission coefficient and wave attenuation constant of the metamaterial pipe are obtained based on the proposed analytical method. In Fig. 5(a), the transmission coefficients are illustrated for the two cases. It is observed that the BS and LR bandgaps can be generated, and the transmission coefficients nearby the LR bandgap are different for the two cases.

Table 2
The natural frequencies (rad/s) under different fluid velocity (Li et al., 2013).

Categories	$v = 0$			$v = 15 \text{ m/s}$			$v = 25 \text{ m/s}$		
	Present	Reference	Error	Present	Reference	Error	Present	Reference	Error
ω_1	4.3718	4.3718	0	4.2858	4.2857	< 0.01%	4.1293	4.1291	< 0.01%
ω_2	17.4761	17.4705	0.03%	17.4000	17.3946	0.03%	17.2643	17.2593	0.03%
ω_3	39.2730	39.2461	0.07%	39.1995	39.1729	0.07%	39.0695	39.0426	0.07%
ω_4	69.7013	69.6165	0.12%	69.6290	69.5445	0.12%	69.5002	69.4163	0.12%
ω_5	108.6727	108.4682	0.19%	108.6011	108.3968	0.19%	108.4735	108.2697	0.19%

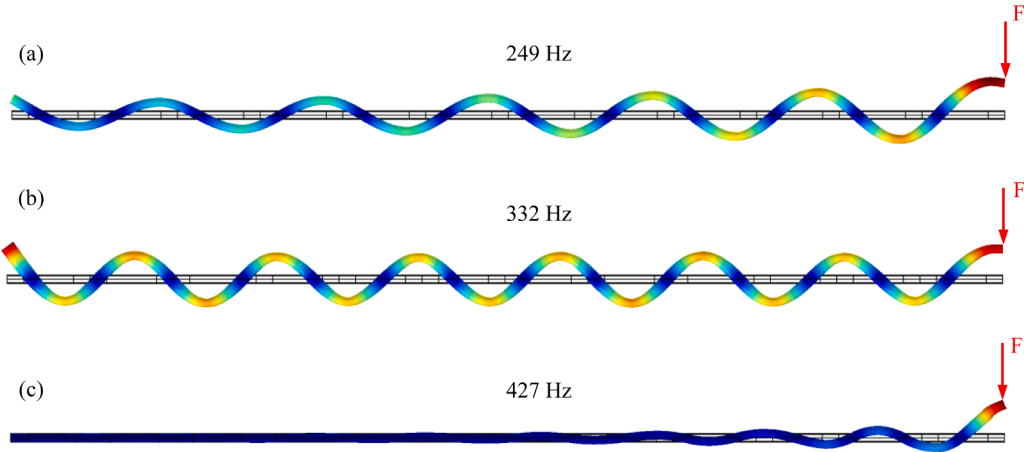


Fig. 3. Vibration distributions of the metamaterial pipe: (a) BS bandgap, (b) pass band, (c) LR bandgap.

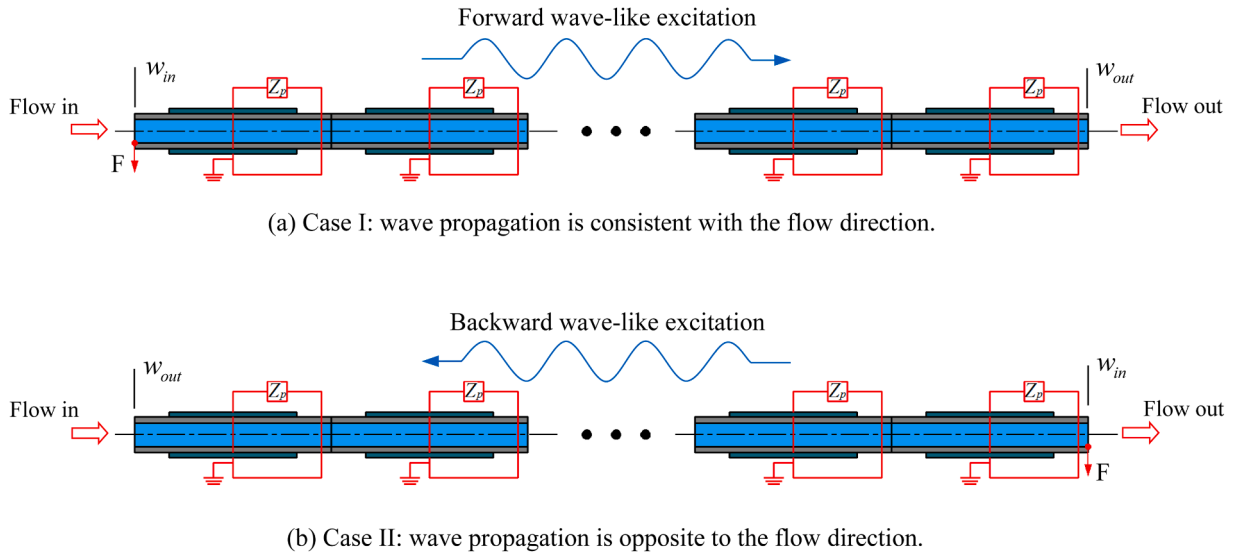


Fig. 4. The propagation directions of wave-like excitations: (a) Case I, (b) Case II.

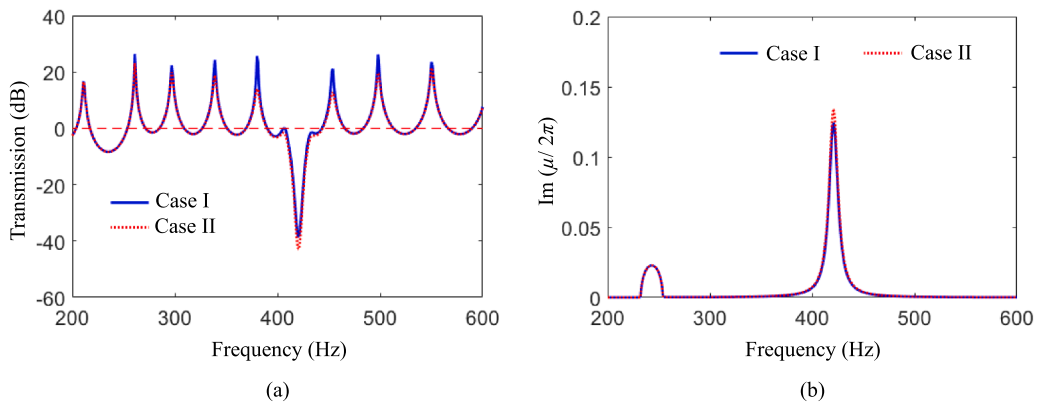


Fig. 5. Bandgap characteristics of Case I and Case II, for $v = 25$ m/s, $L_p = 0.8$ H, $C_p = 100$ nF and $R_p = 50$ Ω : (a) transmission coefficient, (b) attenuation constant.

Specifically, the transmission coefficient near the LR bandgap in Case II is smaller than that in Case I. It indicates that when the wave propagation direction is opposite to the flow direction, the vibration reduction effect can be enhanced within the LR bandgap and the nearby peaks. Therefore, the vibration reduction performance can be adjusted by the internal fluid flow direction, and the flow damping can lead to non-reciprocal wave propagation in metamaterial pipes. This is because the weak non-reciprocal transmission of the bending vibration in metamaterial pipes is generated by mechanical bias, and the mechanical bias is induced by Coriolis force (Trainiti and Ruzzene, 2016; Beli et al., 2015). Due to the non-zero fluid velocity inside the fluid-conveying pipe, the frequency of reflected bending wave is shifted relative to the incident bending wave, which changes the conditions for destructive interference and generates non-reciprocity.

Then in Fig. 5(b), the wave attenuation constant is plotted for the two cases. It can be observed that the BS bandgap remains unchanged in both cases. While the attenuation constants of the forward and backward bending waves exhibit noticeable difference within the LR bandgap. This further proves the non-reciprocal wave propagation in the proposed metamaterial pipe. Therefore, the metamaterial pipes can be regarded as a non-reciprocal system, whose non-reciprocity is induced by the “flow damping” of the internal fluid.

In Fig. 6, the mappings of bandgaps are illustrated as functions of fluid velocity. It can be observed from Fig. 6(a) that for case I and case II, as the fluid velocity increasing from zero to $v = 25$ m/s, the start and end frequencies of the BS bandgap move slightly to lower frequencies while the width of BS bandgap is almost unchanged. Then in Fig. 6(b) and (c), the location and width of LR bandgap for cases I and II are almost not changed as increasing the fluid velocity. It should be emphasized that the vibration reduction performance within LR bandgap will be different for case I and case II. For case I, the vibration reduction performance within LR bandgap is weakened as increasing the flow velocity. For case II, the vibration reduction performance within LR bandgap is enhanced as increasing the flow velocity. Therefore, the influence of flow velocity on vibration reduction performance of LR bandgap should be considered separately for the two cases. It reveals that due to the directionality of fluid flow, the vibration reduction performance within the LR bandgap frequency range can be enhanced or weakened by reversing the direction of fluid flow to be the same or opposite to the direction of vibration propagation.

4.3. Influence of external shunting circuits

The resonant shunt circuits are adopted as the external controllable circuits, so that the bandgap property of the metamaterial pipe

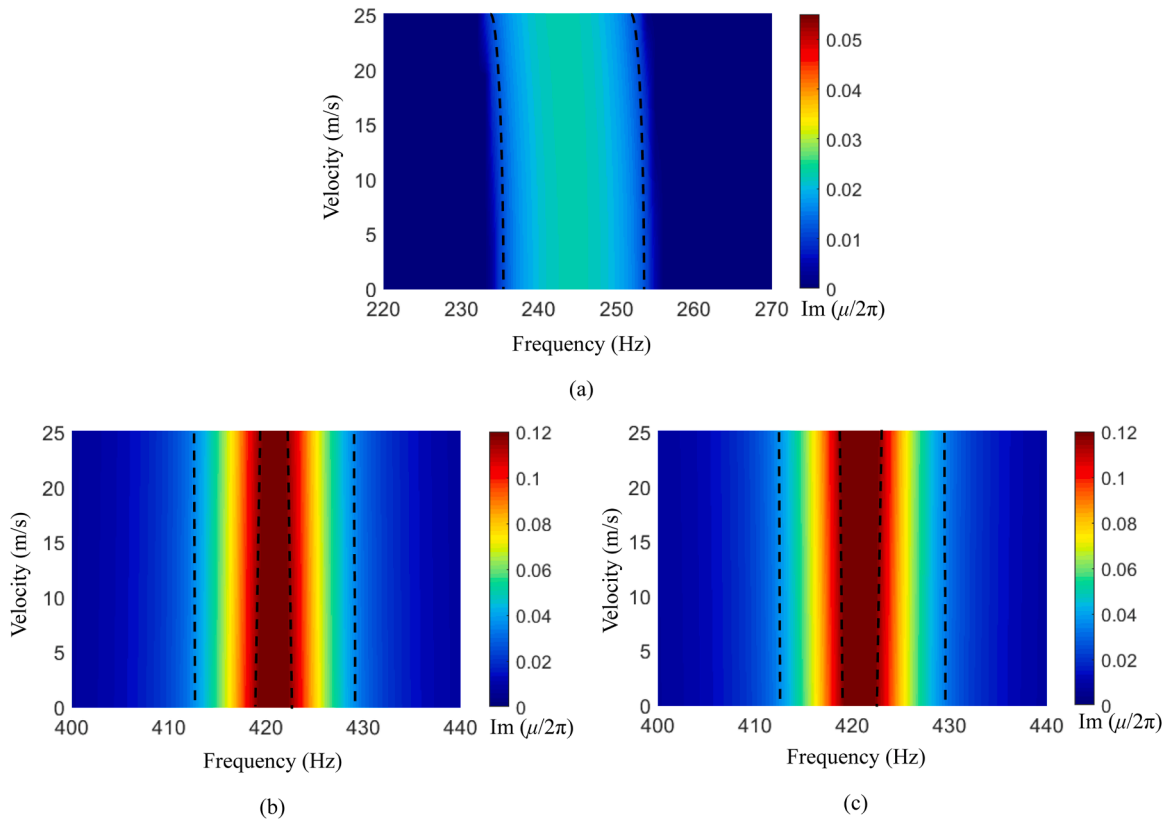


Fig. 6. Mappings of bandgaps as functions of internal fluid velocity: (a) the mapping of BS bandgap for Case I and Case II, (b) the mapping of LR bandgap for Case I, (c) the mapping of LR bandgap for Case II.

can be regulated by adjusting the circuit parameters. This section illustrates the influence of external shunting circuits on the vibration reduction performance of metamaterial pipes. In the calculation, the internal fluid velocity is set to $v = 25$ m/s, and Case I is taken as an example.

Fig. 7 shows the influence of resistance on vibration reduction performance, where the inductance and capacitance are taken as $L_p = 0.8$ H and $C_p = 100$ nF. It is observed that the location and width of BS bandgap are almost unchanged with the resistance. The wider LR bandgap can be obtained for resistance $R_p = 100$ Ω , but the vibration reduction performance within the LR bandgap is weakened. This is because the increase in resistance of the piezoelectric shunt circuit increases the damping of the metamaterial pipe. Accordingly, the attenuation constants are plotted in Fig. 7(b). It can be found that the width of LR bandgap can be enlarged by increasing the resistance from $R_p = 50$ Ω to $R_p = 100$ Ω . The wave attenuation within BS bandgap is not affected by increasing the resistance. Therefore, the LR bandgap of the metamaterial pipe can be adjusted and widened by adjusting the resistance, thereby achieving broadband vibration reduction.

The influence of inductance on the bandgap property of metamaterial pipe is illustrated in Fig. 8. The resistance and capacitance are set to $R_p = 50$ Ω and $C_p = 100$ nF. In Fig. 8(a) and (b), the vibration transmission coefficient and wave attenuation constant are calculated for $L_p = 0.5$ H and $L_p = 1.0$ H. The results demonstrate that the location and width of BS bandgap are almost unaffected by the inductance. The LR bandgap can be achieved in lower frequency range by adopting a bigger value of inductance, but the width of LR bandgap will be reduced.

Fig. 9(a) and (b) illustrate the transmission coefficient and attenuation constant of the metamaterial pipe under different capacitances, where the inductance and resistance are taken as $L_p = 0.8$ H and $R_p = 50$ Ω . It can be observed that the BS bandgap remains almost unchanged for different capacitances. By taking a bigger value of capacitance, the lower frequency range of LR bandgap can be obtained. At the same time, the vibration reduction performance within the LR bandgap is reduced, but the width of the LR bandgap is almost not changed. Therefore, the location of LR bandgap can be tuned by properly designing the inductance and capacitance parameters, and the vibration suppression characteristics of the metamaterial pipe can be achieved in lower frequency range.

4.4. Bandgap coupling characteristics

The bandgap coupling characteristics of the metamaterial pipe are investigated in this section. Case I is taken as an example and the fluid velocity is taken as $v = 25$ m/s. The mappings of bandgaps are illustrated in Fig. 10, as functions of circuit parameters. The influence of resistance on bandgap coupling characteristics is shown in Fig. 10(a), where the inductance and capacitance are taken as $L_p = 2.0$ H and $C_p = 100$ nF. It is observed that the widths of BS and LR bandgaps can be enlarged in both sides as the resistance increases from $R_p = 50$ Ω . When the resistance increases to $R = 335$ Ω , the BS and LR bandgaps will merge into one wider bandgap. As the resistance increases to $R_p = 980$ Ω , the bandwidth will be narrowed, and it reveals that the coupling effect is weakened. Since the resonant circuit is composed of resistance, inductance, and capacitance, when the resistance increases to a very large value, the resonant circuit almost becomes an open circuit, and the LR bandgap generated by the resonant circuit almost disappears.

The influence of inductance on bandgap coupling characteristics is illustrated in Fig. 10(b), where the resistance and capacitance are $R_p = 200$ Ω and $C_p = 100$ nF. It can be found that with the increase of inductance, the wider LR bandgap can be obtained in lower frequency range. Subsequently, the BS and LR bandgaps will be merged into a wider bandgap for the inductance greater than $L_p = 2.12$ H. As the inductance increases to a value bigger than $L_p = 2.64$ H, the merged bandgap will be separated into two bandgaps. Then, the influence of capacitance on bandgap coupling characteristics is displayed in Fig. 10(c). The resistance and inductance are taken as $R_p = 200$ Ω and $L_p = 2.0$ H. It is observed that the BS and LR bandgaps can also be merged to a wider bandgap by designing the capacitance in the range of $[1.08, 1.56]$ nF. The above results indicate that by appropriately adjusting the parameters of the shunt circuits, namely resistance, inductance, and capacitance, the BS and LR bandgaps can be coupled and merged into a wider bandgap, and broadband vibration reduction of the metamaterial pipe can be achieved in the low-frequency range.

To evaluate the vibration reduction performance of the metamaterial pipe, the transmission coefficients under different circuit parameters are further illustrated in Fig. 11. It can be found from Fig. 11(a) that by increasing the resistance from $R_p = 100$ Ω to $R_p =$

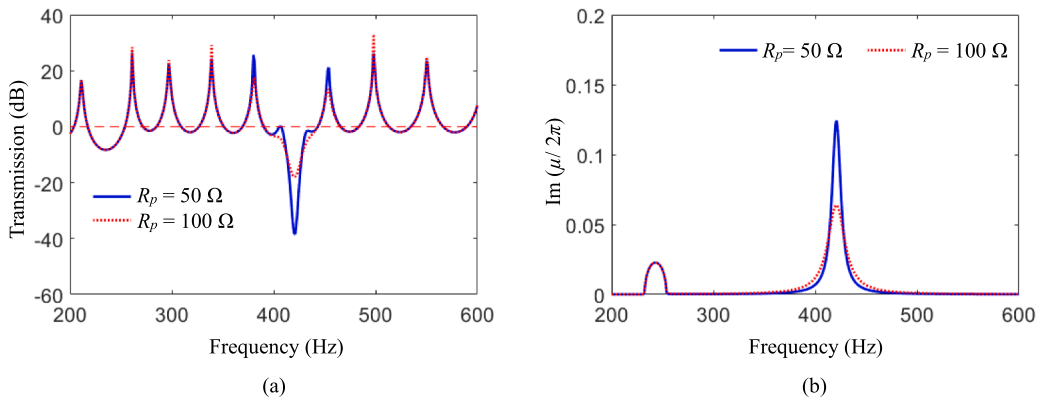


Fig. 7. Influence of resistance R_p on bandgap characteristics for $L_p = 0.8$ H and $C_p = 100$ nF: (a) transmission coefficient, (b) attenuation constant.

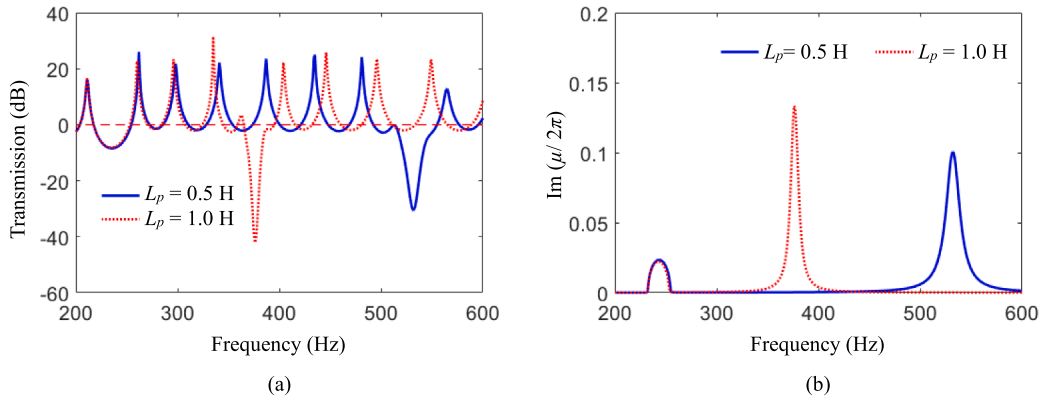


Fig. 8. Influence of inductance L_p on bandgap characteristics for $R_p = 50 \Omega$ and $C_p = 100$ nF: (a) transmission coefficient, (b) attenuation constant.

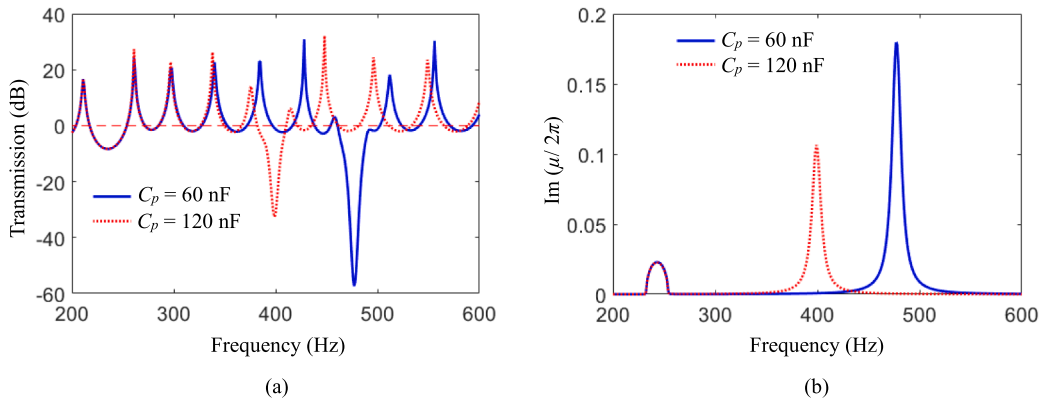


Fig. 9. Influence of capacitance C_p on bandgap characteristics for $L_p = 0.8$ H and $R_p = 50 \Omega$: (a) transmission coefficient, (b) attenuation constant.

600 Ω , a wider coupled bandgap with uniform reduction performance can be obtained for broadband vibration reduction. Fig. 11(b) shows that by increasing the inductance from $L_p = 1.8$ H to $L_p = 2.5$ H, the LR bandgap can be shifted to lower frequency range and coupled with the BS bandgap to form a wider bandgap. It should be noted that the range of inductance in analysis may include large inductance values, which are referred as synthetic inductance. This paper focuses on the regulating effect of inductance on vibration reduction in the low-frequency range, and the larger inductance values are directly adopted in the calculations. In practical applications, the synthetic inductance can be achieved through amplification circuits (Hu et al., 2021). Then, the influence of capacitance on transmission coefficient is plotted in Fig. 11(c). It is observed that as the increase of capacitance from $C_p = 80$ nF to $C_p = 120$ nF, the LR bandgap can couple with the BS bandgap, and the coupled bandgap with uniform reduction performance can be achieved for broadband vibration reduction.

The above analyses on bandgap coupling characteristics demonstrate that broadband vibration reduction with uniform performance can be achieved by properly designing the values of inductance, resistance and capacitance in external shunting circuits. In addition, the circuit parameters can be easily adjusted in practical applications, which is conducive to the design and realization of pipe systems with controllable vibration reduction performance.

5. Conclusions and future work

In this paper, a novel metamaterial pipe is constructed by periodically placing shunted piezoelectric patches on a fluid-conveying pipe. An analytical model is proposed to study the vibration and wave propagation characteristics of the metamaterial pipe, and the calculation accuracy is validated by the results of existing literature and FEM. It is found that the BS and LR bandgaps can be generated for low-frequency vibration reduction. The wider LR bandgap can be achieved in lower frequency range by adjusting the resistance, and the LR bandgap can be tuned by designing inductance and capacitance. The weak non-reciprocal transmission of the bending waves in metamaterial pipe occurs for metamaterial pipe with non-zero fluid velocity, and the bandgap characteristics can be tuned by the direction and velocity of internal fluid flow. The BS and LR bandgaps can be coupled into a wider bandgap by designing the circuit parameters, and broadband vibration reduction with uniform performance of fluid-conveying pipe can be achieved in the low-frequency range.

In summary, a combined analytical model has been proposed for the novel piezoelectric metamaterial pipe, which can effectively

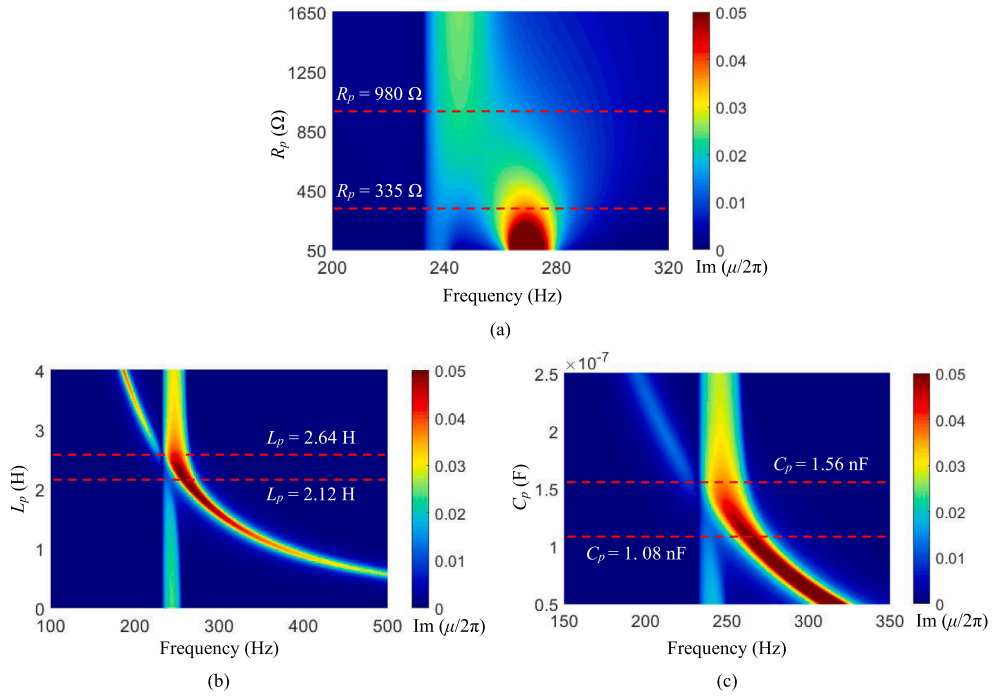


Fig. 10. The mappings of bandgaps as functions of circuit parameters: (a) resistance ($L_p = 2.0 \text{ H}$, $C_p = 100 \text{ nF}$), (b) inductance ($R_p = 200 \Omega$, $C_p = 100 \text{ nF}$), (c) capacitance ($R_p = 200 \Omega$, $L_p = 2.0 \text{ H}$).

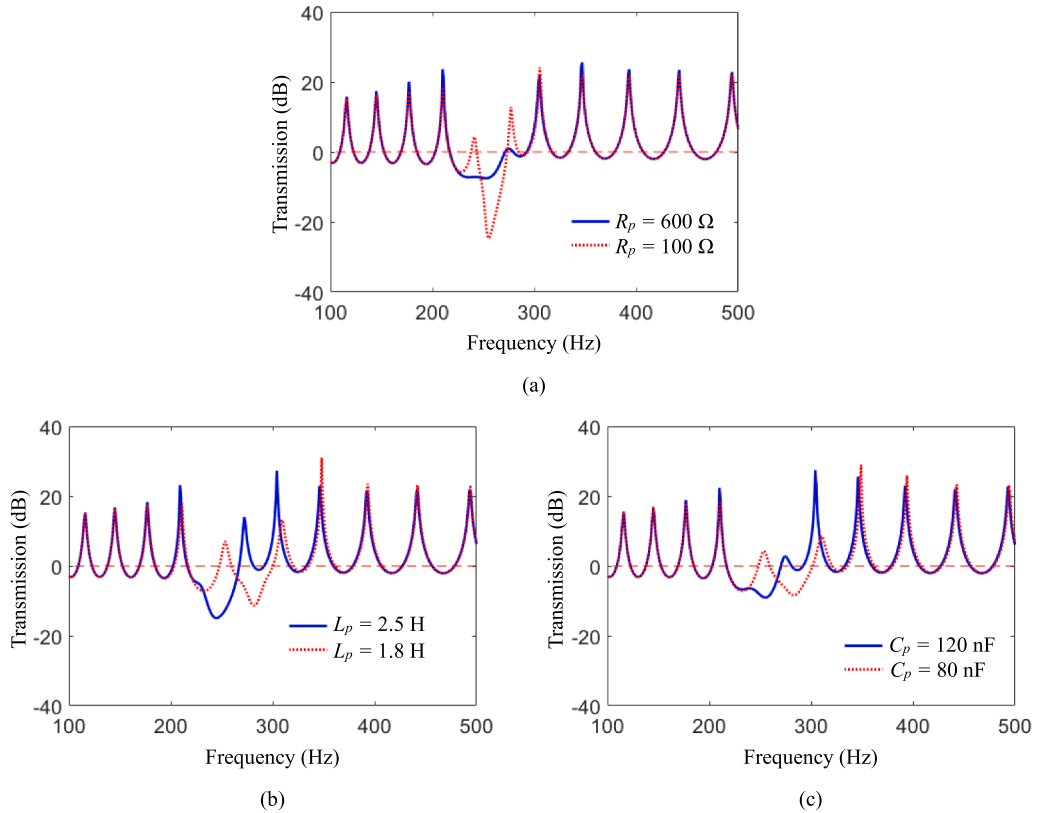


Fig. 11. Transmission coefficients under different circuit parameters: (a) resistance ($L_p = 2.2 \text{ H}$, $C_p = 100 \text{ nF}$), (b) inductance ($R_p = 250 \Omega$, $C_p = 100 \text{ nF}$), (c) capacitance ($R_p = 400 \Omega$, $L_p = 2.0 \text{ H}$).

predict the vibration and wave propagation behaviors. Although the three-dimensional finite element model can prove the accuracy of the analytical model, much experimental research effort is still needed in our future work to extend the findings of the analytical model to practical engineering applications.

CRedit authorship contribution statement

Qifa Lu: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation. **Hongwei Song:** Visualization, Validation, Software, Formal analysis. **Renzhuo Du:** Visualization, Validation, Software, Formal analysis. **Wensheng Ma:** Resources, Project administration, Funding acquisition. **Chunchuan Liu:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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